

Experiment 1.3

Variable Transformation between Cartesian, Cylindrical, and Spherical coordinate systems

Code File Name: Exp1_3_Sph2CartCyl.m

Code

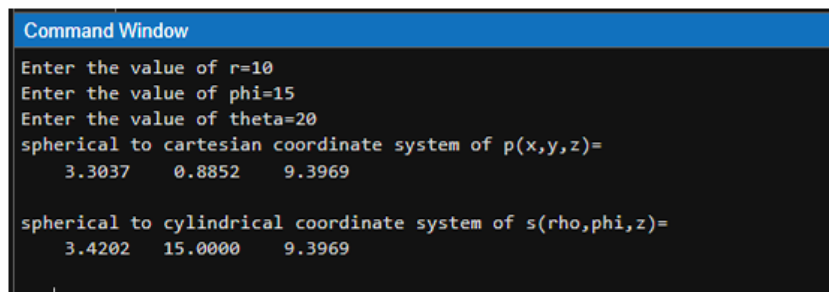
```
clc;
r=input('Enter the value of r=');
phi=input('Enter the value of phi=');
theta=input('Enter the value of theta=');
% Spherical to Cartesian
x=r*sind(theta)*cosd(phi);
y=r*sind(theta)*sind(phi);
z=r*cosd(theta);
disp('spherical to cartesian coordinate system of p(x,y,z)=');
disp([x y z]);
% Spherical to Cylindrical
rho=r*sind(theta);
disp('spherical to cylindrical coordinate system of s(rho,phi,z)=');
disp([rho phi z]);
```

Command Prompt

```
Enter the value of r=10
Enter the value of phi=15
Enter the value of theta=20
spherical to cartesian coordinate system of p(x,y,z)=
    3.3037    0.8852    9.3969

spherical to cylindrical coordinate system of s(rho,phi,z)=
    3.4202   15.0000    9.3969
```

Output Graph / Screenshot

A screenshot of the MATLAB Command Window. The window has a blue title bar that says "Command Window". The background is black with white text. The text shows the user entering values for r, phi, and theta, followed by the program's output for both Cartesian and cylindrical coordinate transformations.

```
Command Window
Enter the value of r=10
Enter the value of phi=15
Enter the value of theta=20
spherical to cartesian coordinate system of p(x,y,z)=
    3.3037    0.8852    9.3969

spherical to cylindrical coordinate system of s(rho,phi,z)=
    3.4202   15.0000    9.3969
```